

Milestone 2: Data Summary for Crime and What Else? Investigating the Predictive Power of Social Factors on Incarceration Rates

Emery Kerani

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Research Question Reminder

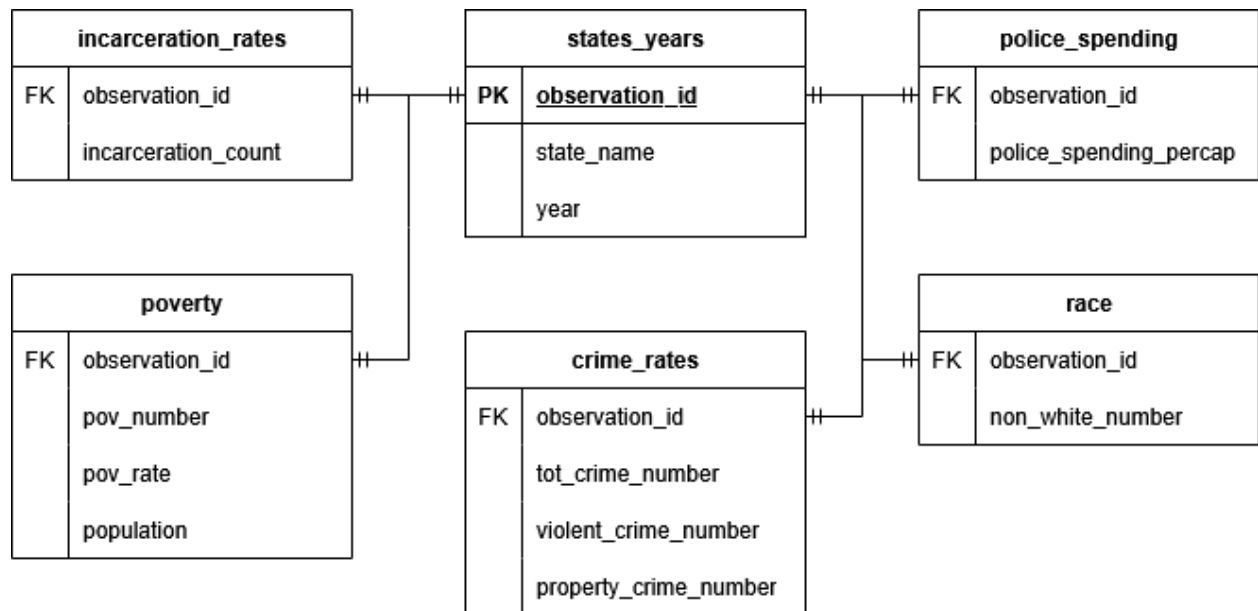
The research questions for this project are “*How do the predictive power of social factors (such as poverty, race, or government spending on police) compare to that of violent and property crime rates when using support vector regression to model the national incarceration rate? And if crime is in fact the best predictor of incarceration rates, which social factor is the best predictor of crime rates?*”. I have sourced and combined data pertaining to both incarceration counts and all predictive factors I’ll be exploring (crime counts, police spending, poverty, and race) for all states between 1980 and 2020; I have a resulting 2,000 state-level observations which I can use to train and test SVR models for each predictive factor. Additionally, all observations have been standardized by calculating rates per 100,000 people, and will be further normalized before modelling in order to prevent any unanticipated influences.

Data inventory

<i>Dataset</i>	<i>Source</i>	<i>Terms</i>	<i>Size</i>	<i>Date range</i>	<i>Refresh policy</i>	<i>Location</i>
Incarceration counts	Corrections Statistical Analysis Tool	Should cite the Bureau of Justice Statistics	2,376 rows	Yearly, 1978-2021	One-time download	data/raw/annual_prison_counts.xlsx
Crime counts	FBI Summary Reporting System	Should cite the FBI SRS	2,389 rows	Yearly, 1979-2024	One-time download	data/raw/estimated_crimes_1979_2024.csv
Police spending	Census Bureau (through Urban Institute)	Should cite both the Census Bureau and Urban Institute	2,030 rows	Yearly, 1980-2022	One-time download	data/raw/police_spending_all.xls
Historical poverty	Census Bureau	Should cite the Census Bureau	2,350 rows	Yearly, 1980-2024	One-time download	data/raw/historical_poverty19.xlsx

Race	IPUMS CPS	Must cite IPUMS and upload to bibliography	12,821, 891 rows (raw)	Yearly, 1980-2020	One-time download	Stored in R/Quarto
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Schema and organization



This diagram represents roughly how the different datasets will be connected, though in lieu of a generated observation ID they will be joined directly based on state and year. In the final version of the cleaned and combined data, raw counts will be replaced with per capita rates per 100,000 people based on the population numbers present in the poverty dataset. The data will be analysis-ready when all of the represented datasets have been joined, there are no null values, and rates have been calculated. The columns in the final dataset are represented in the table below.

Column name	Description	Type	Units
State	The state which an observation corresponds to	str	NA
year	The year an observation corresponds to	int	years
tot_pop	The total population in a given state/year	int	People
incar_rate	The number of people	float	People per 100,000

	incarcerated per 100,000 people in a state/year		
tot_crime_rate	The total number of recorded crimes per 100,000 people in a state/year	float	Crimes per 100,000 people
violent_crime_rate	The number of recorded violent crimes (comprised of homicide, rape, robbery, and aggravated assault) per 100,000 people in a state/year	float	Violent crimes per 100,00 people
prop_crime_rate	The number of recorded property crimes (comprised of burglary, larceny, and motor vehicle theft) per 100,000 people in a state/year	float	Property crimes per 100,000 people
pov_rate	The number of people in poverty per 100,000 in a state/year	float	People per 100,000
non_white_rate	The number of people who identify as anything other than white per 100,000 in a state/year	float	People per 100,000
spending_rate	The number of dollars spent by state and local governments on police per 100,000 people in a state/year	int	Dollars spent per 100,000 people

Ingestion status

All data was pulled manually in .csv or .xlsx format from various online sources (linked in the table above). Step-by-step instructions about how to recreate these pulls can be found in the [data/README.md](#) on Github. With the exception of the IPUMS data (which was too large), all the individual raw files can be found in the [data/raw](#) folder.

Data quality and profiling

- Police spending data was missing for all states in 2001 and 2003 through the website I ended up using. Since the rise in police spending over the years is generally fairly steady, I chose to impute these values by averaging the spending in the years directly

before and after years with missing data (which allowed me to preserve trends by state). This is not a perfect solution; in particular it must be noted that Alaska's spending peaked in 2002, and because of the missingness and imputation I don't have a good idea of what the state's spending around that peak actually looked like.

- For two years (2013 and 2017) of poverty data, there were two estimated counts published for both population and poverty levels due to new measures being introduced in the survey. The new measures were given to only a proportion of the population, but because in theory both versions were getting at the same thing I chose to combine the values by averaging them.
- The poverty and race datasets did not include total counts at a national level; these numbers were calculated by summing state-level numbers for each year of data.
- There are several caveats included in the crime data:
 - Vermont and Nebraska's UCR programs did not provide complete figures for 1997; figures were calculated by applying respective regional trends to 1996 figures.
 - Illinois' state UCR program was unable to provide data about rape from 1995-2013; included figures are based on estimates
 - New York's 2001 figures do not include the deaths resulting from 9/11.
 - Minnesota figures for forcible rape are not included in the data from 2005-2015 due to UCR reporting discrepancies; it's unclear what figures are included, given that data still exists for MN in those years.
 - Several states (Alabama 2011, Montana 2011, New York 2012) changed their reporting practices, meaning that figures may not be comparable to previous years
 - Montana's UCR program could not provide complete figures from 1996-2000; Kentucky's could not provide complete figures from 1996-2003; Wisconsin could not provide complete figures for 1998; and New Hampshire could not provide complete figures in 1998 and 1999. Estimates for these years were generated by applying respective regional trends to previous figures.
 - Kansas was also unable to provide complete figures between 1996 and 2000; in 1996 and 97, totals were extrapolated from January-June state totals, while from 1998-2000 figures were estimated based on regional trends.
 - Maine figures for 1999 are estimates based on the available monthly data from January-October of that year.
 - Florida's 1996 figures are based on aggregated state totals.
 - Delaware was unable to provide forcible rape figures for 1998; these figures were estimated by "reducing the number of reported offenses by the proportion of male forcible rape victims statewide".

Reproducibility.

The script to clean, combine, and preprocess the data for use with SVR can be found in my project's Github repository under [data/processed](#), along with the resulting combined and preprocessed datasets. The R sections were completed in version 4.5.2, and the Python sections were run in version 3.8 through a miniconda environment that included Pandas, Numpy, and Scikit-Learn as requirements.

Ethics and access controls.

As with before, the datasets themselves have few ethics concerns since everything is highly aggregated and cannot be linked back to individuals. I have discovered several caveats during the cleaning process, all of which are listed above and will have to be mentioned as limitations in the final project (along with the fact that both the poverty and race data are based on estimates generated by surveys). It is still the case that all government sources will need to be cited, and the title and citation of the completed project will need to be uploaded to the IPUMS bibliography in accordance with their data use requirements.

No further ingestion work is planned.